

3rd video
why Sample Variance is divided by $n-1$

Sample Variance

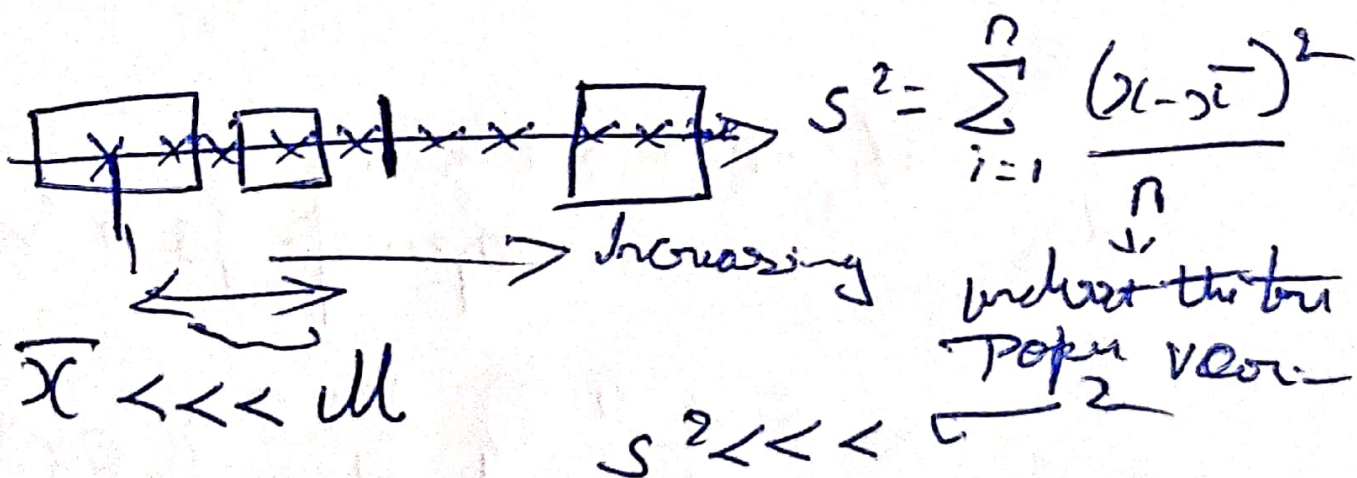
$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$$

$$\sigma^2 = \sum_{i=1}^N \frac{(x_i - \mu)^2}{N}$$

hasil correction ~~***~~ Imp Interview question?

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$$\bar{x} \approx \mu \quad s^2 \approx \sigma^2$$



The diagram shows a horizontal line with several 'x' marks representing data points. A vertical line segment is drawn at the left end, and a horizontal double-headed arrow is drawn below it, indicating a range or spread. To the right of the data points, the formula for sample variance is written: $s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n}$. Below the formula, the text 'increasing' is written with an arrow pointing to the right. To the right of the diagram, the text 'under the true Popu Variance' is written with an arrow pointing to the formula. Below the diagram, the text $\bar{x} \ll \mu$ is written. Below the formula, the text $s^2 \ll \sigma^2$ is written.

$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n}$$

increasing

under the true Popu Variance

$\bar{x} \ll \mu$

$s^2 \ll \sigma^2$

$$D.O.f = n - 1$$

Degree of freedom